PharmaSense AI — Architecture & Findings

Data & Dashboards Submission | Tapabrata Maity | Data Analyst Intern

1. Project Overview

PharmaSense AI is a fictional pharma R&D and clinical operations scenario used as a portfolio/internship project. The full brief describes a multi-agent GenAI assistant; this submission covers the **data and dashboards scope**: loading and validating the provided dataset, answering key business questions in SQL, and building evaluation/observability dashboards. The scope was confirmed with my supervisor as data + dashboards only, with no AI/agent layer required.

2. Data Architecture

Seven linked CSV tables (compounds, clinical_trials, trial_sites, lab_results, adverse_events, research_documents, agent_interaction_logs — 3,700+ records) were loaded into a SQLite database (`pharmasense.db`). SQLite was chosen over a managed warehouse (Snowflake/Postgres) since it needs no server setup and is fully portable for a local, submittable project.

Layer	What it does	Tool used
Data layer	7 structured tables + 1 unstructured text table	SQLite (pharmasense.db)
Analysis layer	Business-question SQL queries, data quality checks	SQL (queries.sql)
Evaluation / observability	Dashboards on agent logs and trial/AE data	Excel (pivot-style formulas + charts)

Trade-off considered: the project guide's reference stack is Dataiku + Snowflake Cortex. That was rejected here in favor of SQLite + Excel because it requires no paid/cloud accounts, is easy to verify end-to-end, and the guide explicitly treats the platform as swappable — the five-layer separation matters more than the specific product.

3. Data Quality Checks

Passed: no duplicate rows or primary keys; no orphan foreign keys across any table relationship; every adverse event's site belongs to its trial; every trial's therapeutic area matches its compound's area; no negative lab values or out-of-range toxicity scores; every trial has at least one site.

Issues found and how they were handled:

Issue	Count	Handling
Actual enrollment > target enrollment	6 trials	Capped % enrolled at 100 in dashboards
Sum of site enrollment ≠ trial actual_enrollment	105 / 110 trials (avg gap 13.7%)	Used trial table for trial totals; site table only for site/country views
Adverse event dated before trial start	182 AEs, 66 trials	Flagged; excluded from time-trend analysis
AEs logged at 'Pending Activation' sites	179 AEs	Flagged as a business-logic inconsistency
Trials overdue (past planned end, still active)	39 trials	Reported as "overdue" rather than dropped
Completed trials with a future planned end date	4 trials	Flagged
Same patient_code in >1 trial	3 codes	Flagged for PII/redaction review
Lab results dated before compound synthesis_date	664 / 2,000	Flagged; excluded from time-based claims

Full detail: `data_quality_report.md`.

4. Key SQL Findings

- Only 1 Phase II oncology trial is below 60% enrollment among in-progress trials (TRL-0037, 40.5%).

- Enrollment rate by phase: Phase I 71.8%, Phase II 75.4%, Phase III 65.8%, Phase IV 85.2%.
- 43 adverse events are classified Serious (all also Severe); 10 had a fatal outcome.
- hERG cardiotoxicity screen failures flag several compounds as cardiotoxicity-risk candidates worth a literature cross-check.
- Lab pass rates range from 73% (cell viability) to 80% (selectivity panel) across experiment types.

Reproducible queries: `queries.sql` (10 core queries, SQLite/DuckDB compatible).

5. Dashboards (Evaluation & Observability)

Built in Excel (`PharmaSense_Dashboards.xlsx`) with formula-driven summary tables (SUMIF/COUNTIFS/AVERAGEIF/MAXIFS) over raw-data Excel Tables, plus native charts — so every number recalculates if the source data changes.

Sheet	Contents
Agent_Summary	Calls, avg latency, avg tokens, avg rating, escalation % by agent; calls/latency by tool; 3 charts
Trial_AE_Summary	Enrollment % by phase, trials by status, AEs by severity, enrollment by country, serious-AE outcomes; 4 charts
Agent_Logs_Raw / Trials_Raw / AE_Raw / Sites_Raw	Raw data as Excel Tables for pivoting or further analysis

Key observation: the Adverse Event Triage agent is the only one with any escalations (18.3% of its 71 calls); every other agent shows 0% — worth flagging as either a true pattern or a possible gap in how the other agents handle escalation-worthy cases.

6. Limitations & Next Steps

- Several date-quality issues (AEs before trial start, labs before synthesis) suggest randomly generated synthetic dates rather than a real data issue — worth a note for whoever reviews this.
- Scope was limited to data + dashboards; the RAG/agent layer (Steps 2-6 of the original guide) was not built.
- At 10x data volume, SQLite would be replaced with a proper warehouse (Postgres/Snowflake) and the Excel dashboards with Tableau or Power BI for better interactivity and refresh automation.

Deliverables: pharmasense.db · queries.sql · data_quality_report.md · PharmaSense_Dashboards.xlsx · this document.